AWS DevOps Engineer Intern Assignment Report

Submitted By

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- **GitHub Repository:** **https://github.com/JaidevCodes/AWS-DevOps-Intern-Assignment**
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1. Objective

The objective of this assignment was to deploy a static website on an AWS EC2 Ubuntu instance using Nginx, perform basic Linux administration tasks, host a custom HTML webpage, manage the project using Git and GitHub, and document the complete deployment process.

2. AWS Services Used

• Amazon EC2: Hosted the Ubuntu server.
• Security Groups: Allowed SSH (22) and HTTP (80).
• Amazon VPC: Default virtual network.
• Internet Gateway: Enabled public internet access.

3. Linux Commands Used

Command	Purpose
sudo apt update	Update package index
sudo apt install nginx -y	Install Nginx
sudo systemctl status nginx	Check Nginx status
sudo systemctl restart nginx	Restart Nginx
df -h	Check disk usage
free -h	Check memory usage
ps aux	View running processes
sudo nano /var/www/html/index.html	Edit webpage
git add .	Stage files
git commit -m "message"	Commit changes
git push origin main	Push to GitHub

4. Website Deployment Process

1. Launched an Ubuntu EC2 instance.
2. Configured a Security Group allowing inbound ports 22 and 80.
3. Connected to the EC2 instance using SSH.
4. Updated packages and installed Nginx.
5. Verified the Nginx service.
6. Created a custom HTML page and replaced the default Nginx page.
7. Verified the website using the EC2 Public IP.
8. Uploaded project files to GitHub.

5. Problems Faced and Resolutions

SSH Login Issue

Initially, I attempted to connect to the EC2 instance using Windows Command Prompt (CMD). After encountering issues, I switched to Windows PowerShell, but the SSH connection still failed due to SSH key handling and environment-related problems. Finally, I used Git Bash, where the SSH connection worked successfully after using the correct private key and command syntax. This helped me understand the differences between Windows terminal environments when working with OpenSSH.

Git Push (Non-Fast-Forward) Error

While pushing my local repository, Git rejected the push because the remote repository already contained commits. I synchronized the local repository with the remote using git pull and then pushed the changes successfully.

Screenshots Folder Missing

I initially forgot to commit and push the screenshots folder. After noticing the issue, I added the folder, committed the changes, and pushed them to GitHub.

6. Learnings

- Provisioning and managing AWS EC2 instances.
- Configuring Security Groups.
- Connecting to Linux servers via SSH.
- Installing and managing Nginx.
- Hosting a static website.
- Using Git and GitHub for version control.
- Troubleshooting SSH and Git issues.

7. Time Taken

Approximately 5 hours.

8. Screenshots

Insert all assignment screenshots in the following order with captions:

1. EC2 Dashboard
2. Security Group
3. SSH Login
4. Nginx Installation
5. Nginx Status
6. Linux Commands
7. Hosted Website
8. GitHub Repository
9. Additional terminal outputs.

9. Conclusion

This assignment was completed successfully by deploying a static website on an AWS EC2 instance, configuring Nginx, performing Linux administration tasks, and managing the project using Git and GitHub. The assignment provided practical experience with AWS, Linux, web hosting, version control, and troubleshooting.